MTH425 Differential Equations

Practice Midterm

First-order DEs, higher-order linear DEs, Laplace transforms | 80 marks

No calculator or other aids. This paper covers the midterm topics that past papers skipped, so it is longer than a real midterm: allow about two hours, or split it into Questions 1 to 4 and 5 to 8. Each question is on its own page with space for your working; the worked solution starts on the next page. Every final answer was checked by substitution.

Question 1 [10 marks]

(a) Find the general solution, and any singular solution, of

$$\frac{dy}{dx} = \frac{xy + 3x - y - 3}{xy - 2x + 4y - 8}.$$

(b) Solve the initial-value problem and give the largest interval I on which the solution is defined:

$$x\frac{dy}{dx} - 4y = x^6 e^x, \qquad y(1) = 1.$$

Solution to Question 1

(a) Classify The right side looks hopeless until you factor numerator and denominator by grouping:

$$xy + 3x - y - 3 = (x - 1)(y + 3), \qquad xy - 2x + 4y - 8 = (x + 4)(y - 2).$$

So $\dfrac{dy}{dx} = \dfrac{x-1}{x+4} \cdot \dfrac{y+3}{y-2}$, which is separable.

Singular solution $y = -3$ makes the right side zero, so the constant function $y = -3$ is a solution. Dividing by $y + 3$ below loses it.

Separate and integrate For $y \neq -3$,

$$\frac{y-2}{y+3}\,dy = \frac{x-1}{x+4}\,dx \quad \Rightarrow \quad \int \left(1 - \frac{5}{y+3}\right) dy = \int \left(1 - \frac{5}{x+4}\right) dx,$$

using long division ($y - 2 = (y + 3) - 5$, and the same for x).

(a) $y - 5\ln\lvert y+3\rvert = x - 5\ln\lvert x+4\rvert + c$, plus the singular solution $y = -3$.

Note Exponentiating gives $(y + 3)^5 = C(x + 4)^5 e^{y-x}$. In that form $C = 0$ recovers $y = -3$, so it is only singular relative to the logarithmic form. Either way, say it.

(b) Classify Linear in y. Standard form, dividing by x:

$$\frac{dy}{dx} - \frac{4}{x}\,y = x^5 e^x.$$

$P(x) = -4/x$ is continuous on $(-\infty, 0)$ and $(0, \infty)$. The initial point $x_0 = 1$ lies in $(0, \infty)$.

Integrating factor $\mu = e^{\int -4/x\,dx} = e^{-4\ln x} = x^{-4}$ on $x > 0$. Then

$$\frac{d}{dx}\left(x^{-4}y\right) = x^{-4} \cdot x^5 e^x = xe^x \quad \Rightarrow \quad x^{-4}y = xe^x - e^x + c.$$

$y(1) = 1$ gives $1 = e - e + c$, so $c = 1$.

(b) $y = x^5 e^x - x^4 e^x + x^4, \quad I = (0, \infty)$.

Check $y' = (x^5 + 5x^4)e^x - (x^4 + 4x^3)e^x + 4x^3$, so $xy' - 4y = x^6 e^x$ after the x^5, x^4 and x^3 terms cancel.

Why not all x The formula is defined everywhere, but the condition at $x_0 = 1$ only pins down c on $(0, \infty)$. On $x < 0$ the general solution is $x^4(xe^x - e^x + k)$ with its own constant k, and because x^4 and its derivative vanish at 0, any k joins smoothly. So the IVP determines y only on the interval containing x_0 where $P = -4/x$ is continuous.

Question 2 [10 marks]

(a) Show that the equation is not exact, find an integrating factor that makes it exact, and solve the initial-value problem:

$$\left(x^2 + y^2 + x\right) dx + xy\, dy = 0, \qquad y(1) = 1.$$

(b) Solve the initial-value problem by a suitable substitution, and state the interval of validity:

$$\frac{dy}{dx} = 1 + e^{y-x+5}, \qquad y(0) = -5.$$

Solution to Question 2

(a) Exactness test $M = x^2 + y^2 + x$, $N = xy$:

$$M_y = 2y, \qquad N_x = y, \qquad M_y \neq N_x,$$

so it is not exact.

Integrating factor Try a factor in x alone:

$$\frac{M_y - N_x}{N} = \frac{2y - y}{xy} = \frac{1}{x},$$

which depends on x only, so $\mu(x) = e^{\int dx/x} = x$. Multiplying through:

$$\left(x^3 + xy^2 + x^2\right) dx + x^2 y \, dy = 0, \qquad M_y = 2xy = N_x. \text{ Exact.}$$

Potential function From $f_y = x^2 y$: $f = \frac{1}{2}x^2y^2 + g(x)$. Then $f_x = xy^2 + g'(x) = x^3 + xy^2 + x^2$, so $g' = x^3 + x^2$ and $g = \frac{1}{4}x^4 + \frac{1}{3}x^3$.

$$\tfrac{1}{4}x^4 + \tfrac{1}{2}x^2y^2 + \tfrac{1}{3}x^3 = c.$$

At $(1, 1)$: $c = \frac{1}{4} + \frac{1}{2} + \frac{1}{3} = \frac{13}{12}$. Multiplying by 12:

(a) $3x^4 + 6x^2y^2 + 4x^3 = 13$

(b) Classify Not separable, linear, exact or Bernoulli as written, but the exponent contains $y - x + 5$, so substitute $u = y - x + 5$. Then $u' = y' - 1$ and the equation becomes

$$\frac{du}{dx} = e^u \quad \Rightarrow \quad e^{-u}\, du = dx \quad \Rightarrow \quad -e^{-u} = x + c.$$

At $x = 0$, $u = -5 - 0 + 5 = 0$, so $-1 = c$. Then $e^{-u} = 1 - x$ and $u = -\ln(1 - x)$, which needs $1 - x > 0$.

(b) $y = x - 5 - \ln(1 - x)$, valid on $(-\infty, 1)$.

Check $y' = 1 + \dfrac{1}{1 - x}$, and $e^{y-x+5} = e^{-\ln(1-x)} = \dfrac{1}{1 - x}$. At $x = 0$: $y = -5$.

Question 3 [10 marks]

(a) Verify that $y_1 = x$ and $y_2 = e^x$ are solutions of

$$(1 - x)y'' + xy' - y = 0.$$

Show that they form a fundamental set of solutions on any interval that doesn't contain $x = 1$, and write the general solution.

(b) Solve the IVP $y(0) = 1$, $y'(0) = 0$ for this equation. On what largest interval does the existence-uniqueness theorem guarantee the solution is the only one?

(c) The general solution of $y'' + 16y = 0$ is $y = c_1 \cos 4x + c_2 \sin 4x$. Solve each boundary-value problem, or explain why its solution isn't unique.

(i) $y(0) = 0,\ y(\pi/2) = 0$ (ii) $y(0) = 0,\ y(\pi/8) = 0$

Solution to Question 3

(a) Verify For $y_1 = x$: $y_1'' = 0$, so $(1-x)(0) + x(1) - x = 0$. For $y_2 = e^x$: $(1-x)e^x + xe^x - e^x = 0$.

Wronskian

$$W(x, e^x) = \begin{vmatrix} x & e^x \\ 1 & e^x \end{vmatrix} = xe^x - e^x = (x-1)e^x.$$

$W \neq 0$ for every $x \neq 1$, so on any interval avoiding $x = 1$ the two solutions are linearly independent and form a fundamental set. By superposition every solution there is a combination of them:

(a) $y = c_1 x + c_2 e^x$ on $(-\infty, 1)$ or $(1, \infty)$.

(b) $y' = c_1 + c_2 e^x$.

$$y(0) = c_2 = 1, \qquad y'(0) = c_1 + c_2 = 0 \quad \Rightarrow \quad c_1 = -1.$$

The theorem needs the coefficients continuous and the leading coefficient $a_2(x) = 1 - x$ nonzero on an interval containing $x_0 = 0$. That fails only at $x = 1$.

(b) $y = e^x - x$; unique on $(-\infty, 1)$.

(c) $y(0) = 0$ gives $c_1 = 0$ in both cases, so $y = c_2 \sin 4x$.

(i) $y(\pi/2) = c_2 \sin 2\pi = 0$ for every c_2. Infinitely many solutions $y = c_2 \sin 4x$.
(ii) $y(\pi/8) = c_2 \sin(\pi/2) = c_2 = 0$. Only the trivial solution $y = 0$.

(c) (i) $y = c_2 \sin 4x$ for any c_2 (not unique); (ii) $y = 0$ only.

Note Boundary conditions don't come with a uniqueness guarantee the way initial conditions do. Here the same equation gives either infinitely many solutions or only the trivial one, depending on where the second boundary sits.

Question 4 [10 marks]

The function $y_1 = x + 1$ is a solution of

$$\left(1 - 2x - x^2\right) y'' + 2(1 + x)y' - 2y = 0.$$

Use reduction of order to find a second solution y_2, and write the general solution on an interval where $1 - 2x - x^2 \neq 0$.

Solution to Question 4

Method Linear, second-order, homogeneous, with variable coefficients and one known solution. That is exactly the setting for reduction of order, $y_2 = y_1 \int \dfrac{e^{-\int P\,dx}}{y_1^2}\,dx$, which needs the equation in standard form.

Standard form Divide by $1 - 2x - x^2$:

$$P(x) = \frac{2(1+x)}{1-2x-x^2}.$$

Exponential factor Let $w = 1 - 2x - x^2$, so $dw = -2(1+x)\,dx$. Then

$$-\int P\,dx = \int \frac{dw}{w} = \ln|w| \quad \Rightarrow \quad e^{-\int P\,dx} = 1 - 2x - x^2,$$

dropping the absolute value. A sign is a constant multiple, and constant multiples don't matter for y_2.

Second solution Write the numerator in terms of $x+1$: $1 - 2x - x^2 = 2 - (x+1)^2$.

$$\begin{aligned} y_2 &= (x+1)\int \frac{2-(x+1)^2}{(x+1)^2}\,dx = (x+1)\int \left(\frac{2}{(x+1)^2} - 1\right) dx \\ &= (x+1)\left(-\frac{2}{x+1} - x\right) = -2 - x - x^2. \end{aligned}$$

Dropping the overall sign, $y_2 = x^2 + x + 2$.

$y_2 = x^2 + x + 2, \qquad y = c_1(x+1) + c_2\left(x^2 + x + 2\right)$

Check With $y_2' = 2x + 1$ and $y_2'' = 2$: $(1 - 2x - x^2)(2) + 2(1+x)(2x+1) - 2(x^2 + x + 2) = (2 - 4x - 2x^2) + (4x^2 + 6x + 2) - (2x^2 + 2x + 4) = 0$. y_2/y_1 is not constant, so the pair is independent.

Question 5 [10 marks]

(a) Find the general solution of

$$y''' + 3y'' - 4y = 0.$$

(b) Solve the initial-value problem

$$y'' + 4y' + 4y = 25\cos x, \qquad y(0) = 0, \quad y'(0) = 0.$$

Solution to Question 5

(a) The auxiliary equation is $m^3 + 3m^2 - 4 = 0$. Any rational root divides 4, and $m = 1$ works: $1 + 3 - 4 = 0$. Dividing by $m - 1$:

$$m^3 + 3m^2 - 4 = (m-1)\left(m^2 + 4m + 4\right) = (m-1)(m+2)^2.$$

A simple root $m = 1$, and $m = -2$ with multiplicity 2.

(a) $y = c_1 e^{x} + c_2 e^{-2x} + c_3 x e^{-2x}$

(b) Complementary solution $m^2 + 4m + 4 = (m+2)^2$, so $y_c = c_1 e^{-2x} + c_2 x e^{-2x}$.

Trial form For $25\cos x$, take $y_p = A\cos x + B\sin x$. Neither term is in y_c, so no change is needed.

$$y_p' = -A\sin x + B\cos x, \qquad y_p'' = -A\cos x - B\sin x.$$

Substituting and collecting:

$$(-A + 4B + 4A)\cos x + (-B - 4A + 4B)\sin x = 25\cos x.$$

$$\begin{aligned} \cos x: &\quad 3A + 4B = 25, \\ \sin x: &\quad -4A + 3B = 0 \quad \Rightarrow \quad B = \tfrac{4}{3}A. \end{aligned}$$

Then $3A + \frac{16}{3}A = \frac{25}{3}A = 25$, so $A = 3$ and $B = 4$.

Initial conditions $y = (c_1 + c_2 x)e^{-2x} + 3\cos x + 4\sin x$.

$$\begin{aligned} y(0) &= c_1 + 3 = 0 && \Rightarrow \quad c_1 = -3, \\ y'(0) &= c_2 - 2c_1 + 4 = 0 && \Rightarrow \quad c_2 = -10. \end{aligned}$$

(b) $y = -(3 + 10x)e^{-2x} + 3\cos x + 4\sin x$

Check At $x = 0$: $y = -3 + 3 = 0$ and $y' = -10 + 6 + 4 = 0$. For $y_p = 3\cos x + 4\sin x$: $y_p'' = -3\cos x - 4\sin x$, $4y_p' = 16\cos x - 12\sin x$ and $4y_p = 12\cos x + 16\sin x$. The cosines add to $(-3 + 16 + 12)\cos x = 25\cos x$, and the sines to $(-4 - 12 + 16)\sin x = 0$.

Question 6 [10 marks]

(a) Find the general solution of

$$x^2y'' + 3xy' + 5y = 0, \qquad x > 0.$$

(b) Use variation of parameters to find the general solution of

$$x^2y'' - xy' + y = 2x, \qquad x > 0.$$

Solution to Question 6

(a) Classify Cauchy-Euler: each x^k matches $y^{(k)}$. Try $y = x^m$:

$$m(m-1) + 3m + 5 = m^2 + 2m + 5 = 0 \quad \Rightarrow \quad m = -1 \pm 2i.$$

A complex pair $\alpha \pm \beta i$ gives $x^\alpha \cos(\beta \ln x)$ and $x^\alpha \sin(\beta \ln x)$.

(a) $y = x^{-1}[c_1 \cos(2 \ln x) + c_2 \sin(2 \ln x)]$

(b) Complementary solution Cauchy-Euler again:

$$m(m-1) - m + 1 = m^2 - 2m + 1 = (m-1)^2 = 0,$$

a repeated root, so $y_1 = x$ and $y_2 = x \ln x$.

Standard form Variation of parameters needs leading coefficient 1. Divide by x^2: $y'' - \frac{1}{x}y' + \frac{1}{x^2}y = \frac{2}{x}$, so $f(x) = 2/x$, not $2x$. Using $2x$ here is the classic mistake, and every later step inherits it.

Wronskian

$$W = \begin{vmatrix} x & x \ln x \\ 1 & \ln x + 1 \end{vmatrix} = x \ln x + x - x \ln x = x.$$

Parameters

$$u_1' = -\frac{y_2 f}{W} = -\frac{x \ln x \cdot (2/x)}{x} = -\frac{2 \ln x}{x} \qquad \Rightarrow \quad u_1 = -(\ln x)^2,$$
$$u_2' = \frac{y_1 f}{W} = \frac{x \cdot (2/x)}{x} = \frac{2}{x} \qquad \Rightarrow \quad u_2 = 2 \ln x.$$

(The first uses $w = \ln x$, $dw = dx/x$.) Then

$$y_p = u_1 y_1 + u_2 y_2 = -x(\ln x)^2 + 2x(\ln x)^2 = x(\ln x)^2.$$

(b) $y = c_1 x + c_2 x \ln x + x(\ln x)^2$

Check $y_p' = (\ln x)^2 + 2 \ln x$ and $y_p'' = (2 \ln x + 2)/x$, so $x^2 y_p'' - x y_p' + y_p = (2x \ln x + 2x) - (x(\ln x)^2 + 2x \ln x) + x(\ln x)^2 = 2x$.

Question 7 [10 marks]

(a) Use the definition $\mathcal{L}\{f(t)\} = \int_0^\infty e^{-st} f(t)\,dt$ to find $\mathcal{L}\{f(t)\}$, where

$$f(t) = \begin{cases} 0, & 0 \le t < 1 \\ t, & t \ge 1 \end{cases}$$

(b) Write $g(t)$ in terms of unit step functions and find $\mathcal{L}\{g(t)\}$:

$$g(t) = \begin{cases} \sin t, & 0 \le t < \pi \\ 0, & t \ge \pi \end{cases}$$

(c) Find $\mathcal{L}\{t^2 e^{-3t} - 4e^{2t}\cos 3t\}$.

Solution to Question 7

(a) f is zero on $[0,1)$, so the integral starts at 1. Integrate by parts with $w = t$, $dv = e^{-st}dt$, taking $s > 0$:

$$\begin{aligned}\mathcal{L}\{f\} &= \int_1^\infty te^{-st}\,dt = \left[-\frac{t}{s}e^{-st}\right]_1^\infty + \frac{1}{s}\int_1^\infty e^{-st}\,dt \\ &= \left(0 + \frac{e^{-s}}{s}\right) + \frac{1}{s}\left[-\frac{1}{s}e^{-st}\right]_1^\infty = \frac{e^{-s}}{s} + \frac{e^{-s}}{s^2}.\end{aligned}$$

The upper limits vanish because $te^{-st} \to 0$ as $t \to \infty$ when $s > 0$.

(a) $\mathcal{L}\{f\} = e^{-s}\left(\frac{1}{s} + \frac{1}{s^2}\right), \quad s > 0$

Cross-check $f = t\mathcal{U}(t-1)$, and the alternative form $\mathcal{L}\{g(t)\mathcal{U}(t-a)\} = e^{-as}\mathcal{L}\{g(t+a)\}$ gives $e^{-s}\mathcal{L}\{t+1\} = e^{-s}(1/s^2 + 1/s)$.

(b) g switches $\sin t$ off at $t = \pi$: $g(t) = \sin t - \sin t\,\mathcal{U}(t-\pi)$. Using $\mathcal{L}\{g(t)\mathcal{U}(t-a)\} = e^{-as}\mathcal{L}\{g(t+a)\}$ with $\sin(t+\pi) = -\sin t$:

$$\mathcal{L}\{\sin t\,\mathcal{U}(t-\pi)\} = e^{-\pi s}\mathcal{L}\{-\sin t\} = -\frac{e^{-\pi s}}{s^2+1}.$$

(b) $g(t) = \sin t - \sin t\,\mathcal{U}(t-\pi), \quad \mathcal{L}\{g\} = \frac{1+e^{-\pi s}}{s^2+1}$

(c) First translation, $\mathcal{L}\{e^{at}f(t)\} = F(s-a)$, applied to $\mathcal{L}\{t^2\} = 2/s^3$ with $a = -3$ and to $\mathcal{L}\{\cos 3t\} = s/(s^2+9)$ with $a = 2$:

(c) $\frac{2}{(s+3)^3} - \frac{4(s-2)}{(s-2)^2+9}$

Note In (c) the shift replaces every s, including the one in the numerator of the cosine transform. Writing $4s/((s-2)^2+9)$ is the usual slip.

Question 8 [10 marks]

Use the Laplace transform to solve each initial-value problem.

(a) $y'' + 9y = e^t, \qquad y(0) = 1, \quad y'(0) = 0$

(b) $y' + y = f(t), \qquad y(0) = 1, \qquad \text{where} \quad f(t) = \begin{cases} 0, & 0 \le t < 1 \\ 5, & t \ge 1 \end{cases}$

Solution to Question 8

(a) Transform Using $\mathcal{L}\{y''\} = s^2Y - sy(0) - y'(0)$ and $\mathcal{L}\{e^t\} = 1/(s-1)$:

$$s^2Y - s + 9Y = \frac{1}{s-1} \quad \Rightarrow \quad Y = \frac{s}{s^2+9} + \frac{1}{(s-1)(s^2+9)}.$$

Partial fractions The quadratic is irreducible, so

$$\frac{1}{(s-1)(s^2+9)} = \frac{A}{s-1} + \frac{Bs+C}{s^2+9}, \qquad 1 = A(s^2+9) + (Bs+C)(s-1).$$

At $s=1$: $1 = 10A$, so $A = \frac{1}{10}$. Matching s^2: $A+B=0$, so $B = -\frac{1}{10}$. Matching constants: $9A - C = 1$, so $C = -\frac{1}{10}$.

$$Y = \frac{s}{s^2+9} + \frac{1}{10}\cdot\frac{1}{s-1} - \frac{1}{10}\cdot\frac{s}{s^2+9} - \frac{1}{30}\cdot\frac{3}{s^2+9}.$$

Invert Pairs used: $1/(s-a) \to e^{at}$, $s/(s^2+k^2) \to \cos kt$, $k/(s^2+k^2) \to \sin kt$.

(a) $y = \frac{9}{10}\cos 3t - \frac{1}{30}\sin 3t + \frac{1}{10}e^t$

Check $y(0) = \frac{9}{10} + \frac{1}{10} = 1$ and $y'(0) = -\frac{1}{10} + \frac{1}{10} = 0$. The trig terms solve $y'' + 9y = 0$, and $\frac{1}{10}e^t$ gives $\frac{1}{10}e^t + \frac{9}{10}e^t = e^t$.

(b) Transform $f(t) = 5\mathcal{U}(t-1)$ and $\mathcal{L}\{\mathcal{U}(t-a)\} = e^{-as}/s$:

$$sY - 1 + Y = \frac{5e^{-s}}{s} \quad \Rightarrow \quad Y = \frac{1}{s+1} + 5e^{-s}\cdot\frac{1}{s(s+1)}.$$

Invert $\dfrac{1}{s(s+1)} = \dfrac{1}{s} - \dfrac{1}{s+1} \to 1 - e^{-t}$. The second translation theorem then delays that by one unit.

(b) $y = e^{-t} + 5\left(1 - e^{-(t-1)}\right)\mathcal{U}(t-1)$, that is, $y = \begin{cases} e^{-t}, & 0 \le t < 1 \\ e^{-t} + 5 - 5e^{-(t-1)}, & t \ge 1 \end{cases}$

Check For $t < 1$, $y' + y = 0$. For $t \ge 1$, $y' + y = 5$. Both pieces equal e^{-1} at $t = 1$, so y is continuous, as the solution of a first-order equation with a jump in f must be.